Bangladesh University of Engineering and Technology

Department of Computer Science and Engineering

Course: CSE 206
Digital Logic Design Sessional
Experiment No. 2

Topic: Truth tables and simplification using Boolean Algebra.

Design and implement the following problems:

Reduce the following Boolean functions in the following list and implement the reduced circuit using simple gates only. You must use the 7400-series IC chips: IC 74x04, IC 74x08, and IC 74x32 corresponding to the logic gates. Make sure to use a minimum no. of logic gates. You must submit both the simplification steps and the circuit.

a. AB'C+AB'D+B+B'C'D'+AB
b. A+A'BC+A'B'D'+A'DC+ABC
c. ABD+A'BC'+BC'D'+B'+C'

Lab sheet: For each of the problems/questions, your lab sheet should cover:

- Required IC chips
- Truth table
- Simplification steps
- Circuit diagram

Answer the following question:

1. What are the key challenges you face when you simplify a circuit using Boolean algebra?
2. Please propose an algorithmic idea to systematize Boolean circuit simplification so that you can devise a program for circuit simplification. Your idea must be different from any existing approaches. It is not expected that your idea would be perfect.
3. If you have a Boolean circuit with 100 variables and 50 terms (SOP), how many possible options for minimal simplification of this circuit?